# Project Synopsis

## 1. Title

## This page gives a quick overview of the project and team.

**Working Project Title**
StudentPulse – Learning Intelligence Dashboard
(An interactive web-based student performance estimation and study-planning workspace: https://studentpulsee.netlify.app)

### Team Details

Details of all four team members are given below. The first student is the Team Lead.

| Student Name | College Name | Branch | Mobile No | Mail ID | Role (Lead/Member) |
|---|---|---|---|---|---|
| Khyati Agarwal | Biyani Girls College | Vidhyadhar Nagar | 7300204880 | khyatiagarwal0751@gmail.com | Team Lead |
| Tanisha Soni | Biyani Girls College | Vidhyadhar Nagar | 7877076578 | tanishasoni3984@gmail.com | Team Member |
| Kumkum Sharma | Biyani Girls College | Vidhyadhar Nagar | 7597490551 | kumkumsharmajpr2006@gmail.com | Team Member |
| Neha Solankee | Biyani Girls College | Vidhyadhar Nagar | 8233602320 | Nehaa.solankii99@gmail.com | Team Member |

### Program Track and Domain

The Program Track and Domain selected for this project are given below.

| Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke) | Domain (Analytics / Machine Learning / Deep Learning / GenAI) |
|---|---|
| Science Spoke | Analytics |

### Guide Details

Guide Name and Designation: (to be filled in by the team)

## 2. Problem Statement

Students often have access to their own study habits and marks, but no simple way to see how routine factors such as study hours, attendance, previous exam marks, assignment completion and sleep may relate to their overall performance. StudentPulse is a student project website that turns these study habits into an interactive learning snapshot with suggested study actions.

The problem statement can be summarised as follows:

- Background: Learning habits such as study time, attendance, revision, assignments and sleep influence how students perform, but they are rarely reviewed together in one place.
- Reason for choosing the problem: The team wanted to give students a friendly, low-pressure way to reflect on their learning and plan what to practise next.
- Importance in real life: Regular reflection and planning help students build consistent routines; the website encourages this with the message that effort matters more than one score.
- Gap: Students lack a single, easy tool that combines a score estimate, a weekly study plan, a focus timer, subject tracking, notes and daily check-ins.
- What the project plans to solve: StudentPulse provides an interactive dashboard, study planner and Focus Lab so students can explore scenarios and turn suggestions into small, practical actions.

Important note: the performance score shown by StudentPulse is an illustrative estimate calculated from the user's inputs using a transparent scoring formula. It is not a trained or validated machine-learning model, and it must not be used for grading, admissions or any other consequential decision.

## 3. Objectives

The specific goals of the project are: (1) to build a web-based dashboard that gives an illustrative performance score estimate from inputs such as study hours, attendance, previous exam score, assignments completed, sleep, class participation and weekly revision sessions; (2) to show the distance between the estimate and the student's target score and to compare a few what-if scenarios; (3) to provide personalised next-step suggestions and a weekly study planner checklist; (4) to provide a Focus Lab with a focus/Pomodoro timer, subject tracker, study notes and daily check-ins; (5) to save check-ins and prediction history locally in the browser and allow them to be exported as a CSV file; and (6) to clearly communicate the limitations of the estimate and keep user data private on the device.

## 4. Methodology / Implementation Plan

The project is carried out as a browser-based web application. The student enters a learning profile on the dashboard, the website calculates an illustrative score estimate, and the result is used to suggest next steps and a weekly plan.

The main parts of the plan are:

- Overall workflow: requirement understanding and interface design; building the Dashboard (inputs, Calculate estimate, Reset inputs, score estimate, distance to target, what-if comparison, action plan); building the Study planner (weekly checklist with progress); building the Focus Lab (focus timer with selectable focus and break minutes, subject tracker, study notes, daily check-in); building Prediction history with local saving and CSV export; and testing and deployment of the website.
- Scoring approach: a transparent, illustrative scoring formula is applied to the user's inputs. No machine-learning algorithm or trained model is used, and the formula has not been validated against real exam results.
- Tools/techniques: web technologies for the browser-based interface, browser local storage for saving check-ins, notes, checklist progress and history on the user's device (nothing is sent to a server), CSV export for history, light/dark appearance option, and hosting on Netlify (https://studentpulsee.netlify.app).

Detailed coding is not included in this synopsis.

## 5. Resources and Timeline

**A. Hardware & Software Requirements**
Requirements for developing and using StudentPulse:

- Laptop/PC or mobile device capable of running a modern web browser (specification to be filled in by the team)
- OS: any operating system that supports a modern web browser
- Programming languages: standard web technologies used to build the browser-based website (to be confirmed by the team)
- Tools: web browser, code/text editor, Netlify for hosting the website

**B. Dataset**
Dataset details:

- Dataset availability: no external dataset is used. The website works on values entered by the user (study hours, attendance, previous exam score, assignments, sleep, participation, revision sessions and target score).
- Source link: not applicable.

- Data collection: user inputs, check-ins, notes and history are stored locally in the user's browser and can be exported as a CSV file. No training data is collected, since no machine-learning model is trained.

**C. Project Timeline**
Tentative phases of the project (dates to be filled in by the team):

- Phase 1 – Planning and design: finalise features and layout of the dashboard. Tentative Completion Date: ____10th Oct 2026_____
- Phase 2 – Dashboard and study planner: inputs, score estimate, what-if comparison, action plan and weekly checklist. Tentative Completion Date: _____12th Oct,2026____
- Phase 3 – Focus Lab and history: focus timer, subject tracker, notes, daily check-in, local history and CSV export. Tentative Completion Date: _13th Oct 2026_____
- Phase 4 – Testing, deployment and documentation: testing the website, hosting on Netlify and final report. Tentative Completion Date: _____13th Oct 2026____

## 6. Expected Outcomes / Conclusion

By the end of the project, the team expects to deliver a working, publicly accessible website that helps students reflect on their learning habits.

Expected outcomes:

- A working demo: the StudentPulse website at https://studentpulsee.netlify.app
- A dashboard giving an illustrative performance score estimate, distance to target and what-if scenario comparison
- A study planner with a weekly checklist and personalised next-step suggestions
- A Focus Lab with a focus timer, subject tracker, study notes and daily check-in
- Locally stored prediction history with CSV export, and a clear statement that the score is an illustrative estimate and not a validated machine-learning prediction

## 7. References

Material referred to for this project:


- Pomodoro Technique (Francesco Cirillo), the focus and break timing method used for the Focus Lab timer
- Study methods used in the study habits section: active recall and spaced revision
- Netlify documentation: https://docs.netlify.com